

□ સમજૂતી :

❖ અગત્યના સૂત્રો :

$$\Rightarrow (a + b)^2 = a^2 + 2ab + b^2$$

$$\Rightarrow (a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$$

$$\Rightarrow (a + b)^2 - (a - b)^2 = 4ab$$

$$\Rightarrow a^2 + b^2 = (a + b)^2 - 2ab$$

$$\Rightarrow a^2 - b^2 = (a + b)(a - b)$$

$$\Rightarrow a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

(અથવા)

$$(a + b)^3 - 3ab(a + b)$$

$$\Rightarrow a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

અથવા

$$(a - b)^3 + 3ab(a + b)$$

$$\Rightarrow (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$\Rightarrow (a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$\Rightarrow (a - b - c)^2 = a^2 + b^2 + c^2 - 2ab + 2bc - 2ac$$

$$\Rightarrow a^4 - b^4 = (a - b)(a + b)(a^2 + b^2)$$

$$\Rightarrow \frac{(a + b)^2 + (a - b)^2}{a^2 + b^2} = 2$$

$$\Rightarrow \frac{(a + b)^2 - (a - b)^2}{ab} = 4$$

$$\Rightarrow a^3 + b^3 + c^3 - 3ab = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

Type # 1

(1) જો $a + b = 5$ અને $ab = 3$ હોય તો $a^3 + b^3$ નું મૂલ્ય જણાવો.

(A) 70 (B) 75 (C) 80 (D) 65

Solution :

$$\Rightarrow (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(5)^3 = a^3 + b^3 + 3 \times 3 \times 5$$

$$125 = a^3 + b^3 + 45$$

$$a^3 + b^3 = 125 - 45$$

$$a^3 + b^3 = 80$$

$$\therefore (C) 80$$

(2) જો $(a - b) = 4$ અને $ab = 2$ હોય તો $a^3 - b^3$ નું મૂલ્ય જણાવો.

(A) 92 (B) 84 (C) 88 (D) 80

Solution :

$$\Rightarrow (a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$4^3 = a^3 - b^3 - 3 \times 2 \times 4$$

$$64 = a^3 - b^3 - 24$$

$$a^3 - b^3 = 64 + 24$$

$$a^3 - b^3 = 88$$

$$\therefore (C) 88$$

(3) જો $x + y = 4$ અને $xy = 6$ હોય તો $\frac{x^2 - y^2}{x^3 - y^3}$ નું મૂલ્ય જણાવો.

(A) $\frac{2}{5}$ (B) $\frac{3}{5}$ (C) $\frac{5}{6}$ (D) $\frac{1}{6}$

Solution :

$$\begin{aligned} \Rightarrow \frac{x^2 - y^2}{x^3 - y^3} &= \frac{\cancel{(x - y)}(x + y)}{\cancel{(x - y)}(x^2 + y^2 + xy)} \\ &= \frac{x + y}{x^2 + y^2 + xy + 2xy - 2xy} \\ &= \frac{x + y}{(x + y)^2 - xy} \\ &= \frac{4}{(4)^2 - 6} \\ &= \frac{4}{16 - 6} \\ &= \frac{4}{10} \\ &= \frac{2}{5} \end{aligned}$$

$$\therefore (A) \frac{2}{5}$$

(4) $1 - 2xy - (x^2 + y^2)$ ના અવયવો નીચેનામાંથી જણાવો.

$$(A) (1 + x - y)(1 + x + y)$$

$$(B) (1 + x - y)(1 - x + y)$$

$$(C) (1 + x + y)(1 - x + y)$$

$$(D) (1 + x + y)(1 - x - y)$$

Solution :

$$\begin{aligned} \Rightarrow 1 - 2xy - (x^2 + y^2) &= 1 - (x^2 + y^2 + 2xy) \\ &= 1 - (x + y)^2 \\ &= (1)^2 - (x + y)^2 \\ (\because a^2 - b^2 &= (a - b)(a + b)) \\ &= (1 - x - y)(1 + x + y) \end{aligned}$$

$$\therefore (D) (1 + x + y)(1 - x - y)$$

(5) $3^{2x-16} \times 4^{x-5} = 64$ હોય, તો $x = ?$

- (A) 6 (B) 8
(C) 4 (D) 9

Solution :

$$\begin{aligned}\Rightarrow 3^{2x-16} \times 4^{x-5} &= 64 \\ 3^{2x-16} \times 4^{x-5} &= 3^0 \times 4^3 \\ 3^{2x-16} &= 3^0 \text{ અને } 4^{x-5} = 4^3 \\ 2x-16 &= 0 \quad x-5 = 3 \\ 2x &= 16 \quad x = 8 \\ x &= \frac{16}{2} \\ x &= 8 \\ \therefore (B) 8\end{aligned}$$

(6) સાદુરૂપ આપો : $\frac{(x^a+b)^4 \times (x^b+c)^4 \times (x^c+a)^4}{(x^a \times x^b \times x^c)^4}$

(A) 1 (B) x^4
(C) 0 (D) $x^{4(a+b+c)}$

Solution :

$$\begin{aligned}\Rightarrow \frac{(x^{a+b})^4 \times (x^{b+c})^4 \times (x^{c+a})^4}{(x^a \times x^b \times x^c)^4} \\ = \frac{x^{4a+4b} \times x^{4b+4c} \times x^{4c+4a}}{x^{4a} \times x^{4b} \times x^{4c}} \\ = \frac{x^{4a+4b+4b+4c+4c+4a}}{x^{4a+4b+4c}} \\ = \frac{x^{8a+8b+8c}}{x^{4a+4b+4c}} \\ = \frac{x^{8(a+b+c)}}{x^{4(a+b+c)}} \\ = x^{4(a+b+c)} \\ \therefore (D) x^{4(a+b+c)}\end{aligned}$$

$\frac{x^m}{x^n} = x^{m-n}$

(7) જો $a + b = 4$ અને $ab = -21$ હોય તો $a^3 + b^3 = ?$

(A) 370 (B) 158
(C) 185 (D) 316

Solution :

$$\begin{aligned}\Rightarrow (a+b)^3 &= a^3 + b^3 + 3ab(a+b) \\ (4)^3 &= a^3 + b^3 + 3 \times (-21) \times 4 \\ 64 &= a^3 + b^3 - 252\end{aligned}$$

$$a^3 + b^3 = 252 + 64$$

$$a^3 + b^3 = 316$$

$$\therefore (D) 316$$

(8) $3x^2 + 2y - 5z$ અને $7x^2 - 3y + 6z$ નો સરવાળો કેટલો થશે ?

- (A) $10x^2 + y + z$ (B) $4x^2 + y + z$
(C) $4x^2 - y + z$ (D) $10x^2 - y + z$

Solution :

$$\begin{aligned}\Rightarrow 3x^2 + 2y - 5z \\ + 7x^2 - 3y + 6z \\ \hline 10x^2 - y + z \\ \therefore (D) 10x^2 - y + z\end{aligned}$$

(9) $7a^3 - 2b^2 + 9c$ અને $-4b^2 + 8c$ નો સરવાળો કેટલો થશે ?

(A) $7a^3 - 6b^2 + 17c$ (B) $7a^3 + 6b^2 + 17c$
(C) $11a^3 - 6b^2 + 17c$ (D) $11a^3 + 6b^2 + 17c$

Solution :

$$\begin{aligned}\Rightarrow 7a^3 - 2b^2 + 9c \\ + 0a^3 - 4b^2 + 8c \\ \hline 7a^3 - 6b^2 + 17c \\ \therefore (A) 7a^3 - 6b^2 + 17c\end{aligned}$$

(10) $12a - 5b, 3a + 6b, -2a - 3b$ નો સરવાળો કેટલો થશે ?

(A) $13a + 2b$ (B) $13a - 2b$
(C) $16a + 8b$ (D) $11a + 9b$

Solution :

$$\begin{aligned}\Rightarrow 12a - 5b + 3a + 6b + (-2a - 3b) \\ 12a - 5b + 3a + 6b - 2a - 3b \\ 13a - 2b \\ \therefore (B) 13a - 2b\end{aligned}$$

(11) $(2x + y)^2 = \underline{\hspace{2cm}}$

- (A) $4x^2 - 4xy + y^2$ (B) $2x^2 - 2xy + y^2$
(C) $4x^2 + 4xy + y^2$ (D) $2x^2 + 2xy + y^2$

Solution :

$$\begin{aligned}\Rightarrow (a+b)^2 &= a^2 + 2ab + b^2 \text{ સૂત્ર મુજબ} \\ (2x+y)^2 &= [2x]^2 + 2 \times 2x \times y + y^2 \\ &= 4x^2 + 4xy + y^2 \\ \therefore (C) 4x^2 + 4xy + y^2\end{aligned}$$

(12) $(x + 5)^2 = ?$

- (A) $x^2 - 10x - 25$ (B) $x^2 + 8x + 10$
(C) $x^2 - 8x + 10$ (D) $x^2 + 10x + 25$

Solution :

$$\Rightarrow (a + b)^2 = a^2 + 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$\Rightarrow (x + 5)^2 = x^2 + 2 \times x \times 5 + 5^2 \\ = x^2 + 10x + 25$$

$$\therefore \text{(D) } x^2 + 10x + 25$$

(13) $(3x + 2)^2 = ?$

- (A) $9x^2 + 12x + 4$ (B) $3x^2 + 6x + 4$
(C) $3x^2 - 6x - 4$ (D) $9x^2 - 12x - 4$

Solution :

$$\Rightarrow (a + b)^2 = a^2 + 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$(3x + 2)^2 = [3x]^2 + 2 \times 3x \times 2 + 2^2 \\ = 9x^2 + 12x + 4$$

$$\therefore \text{(A) } 9x^2 + 12x + 4$$

(14) $(8x + 7)^2 = ?$

- (A) $8x^2 + 56x + 49$ (B) $64x^2 + 112x + 49$
(C) $64x^2 - 112x - 49$ (D) $8x^2 - 56x - 49$

Solution :

$$\Rightarrow (a + b)^2 = a^2 + 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$(8x + 7)^2 = [8x]^2 + 2 \times 8x \times 7 + 7^2 \\ = 64x^2 + 112x + 49$$

$$\therefore \text{(B) } 64x^2 + 112x + 49$$

(15) $(3x - 2)^2 = ?$

- (A) $9x^2 - 12x - 4$ (B) $3x^2 - 6x + 4$
(C) $3x^2 + 6x + 4$ (D) $9x^2 - 12x + 4$

Solution :

$$\Rightarrow (a - b)^2 = a^2 - 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$(3x - 2)^2 = [3x]^2 - 2 \times 3x \times 2 + 2^2 \\ = 9x^2 - 12x + 4$$

$$\therefore \text{(D) } 9x^2 - 12x + 4$$

(16) $(5x - 3)^2 = ?$

- (A) $25x^2 - 30x - 9$ (B) $25x^2 - 30x + 9$
(C) $20x^2 + 10x + 8$ (D) $20x^2 - 10x + 8$

Solution :

$$\Rightarrow (a - b)^2 = a^2 - 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$(5x - 3)^2 = [5x]^2 - 2 \times 5x \times 3 + 3^2 \\ = 25x^2 - 30x + 9$$

$$\therefore \text{(B) } 25x^2 - 30x + 9$$

(17) $(x - 8)^2 = ?$

- (A) $x^2 + 16x + 64$ (B) $x^2 - 32x + 64$
(C) $x^2 - 16x + 64$ (D) $x^2 + 32x + 64$

Solution :

$$\Rightarrow (a - b)^2 = a^2 - 2ab + b^2 \text{ (ફૂલ પુજબ)}$$

$$(x - 8)^2 = [x]^2 - 2 \times x \times 8 + 8^2 \\ = x^2 - 16x + 64$$

$$\therefore \text{(C) } x^2 - 16x + 64$$

(18) $[80]^2 - [10]^2 = ?$

- (A) 6300 (B) 6400 (C) 6500 (D) 6100

Solution :

$$\Rightarrow a^2 - b^2 = (a + b)(a - b) \text{ (ફૂલ પુજબ)}$$

$$80^2 - 10^2 = (80 + 10)(80 - 10) \\ = 70 \times 90 \\ = 6300$$

$$\therefore \text{(A) } 6300$$

(19) $\frac{x^2 - 4}{(x - 2)} = ?$

- (A) $x + 2$ (B) $x + 4$ (C) $x - 2$ (D) $x - 4$

Solution :

$$\Rightarrow a^2 - b^2 = (a + b)(a - b) \text{ (ફૂલ પુજબ)}$$

$$\Rightarrow \frac{x^2 - 4}{x - 2} = \frac{x^2 - 2^2}{x - 2} \\ = \frac{(x - 2)(x + 2)}{(x - 2)} \\ = x + 2$$

$$\therefore \text{(A) } x + 2$$

(20) $\frac{x^2 - 9}{x + 3} = ?$

- (A) $x + 3$ (B) $x - 3$ (C) $x - 9$ (D) $x + 9$

Solution :

$$\Rightarrow a^2 - b^2 = (a + b)(a - b) \text{ (ફૂલ પુજબ)}$$

$$\Rightarrow \frac{x^2 - 9}{x + 3} = \frac{x^2 - 3^2}{x + 3} \\ = \frac{(x - 3)(x + 3)}{(x + 3)} \\ = x - 3$$

$$\therefore \text{(B) } x - 3$$

(21) $(x+y)^2 - (x-y)^2 = ?$

- (A) 4 (B) $4xy$
(C) $x^2 + y^2$ (D) $x^2 - y^2$

Solution :

$$\begin{aligned}\Rightarrow (x+y)^2 - (x-y)^2 \\ = x^2 + 2xy + y^2 - (x^2 - 2xy + y^2) \\ = x^2 + 2xy + y^2 - x^2 + 2xy - y^2 \\ = 4xy\end{aligned}$$

$\therefore$ (B) $4xy$

(22) $(x+y)^2 + (x-y)^2 = ?$

- (A) $2(x^2 - y^2)$ (B) $4xy$
(C) $2(x^2 + y^2)$ (D) $3xy$

Solution :

$$\begin{aligned}\Rightarrow (x+y)^2 + (x-y)^2 \\ \Rightarrow x^2 + \cancel{2xy} + y^2 + x^2 - \cancel{2xy} + y^2 \\ = 2x^2 + 2y^2 \\ = 2(x^2 + y^2)\end{aligned}$$

$\therefore$ (C) $2(x^2 + y^2)$

(23) $\frac{(2.8)^3 + (1.7)^3 + (1.5)^3 - 3 \times 2.8 \times 1.7 \times 1.5}{2.8^2 + 1.7^2 + 1.5^2 - 2.8 \times 1.7 - 1.7 \times 1.5 - 2.8 \times 1.5}$

- (A) 6 (B) 5
(C) 7 (D) 8

Solution :

$$\begin{aligned}\Rightarrow a^3 + b^3 + c^3 - 3abc \\ = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \text{ પુજા} \\ = \frac{(2.8+1.7+1.5)(\cancel{2.8^2+1.7^2+1.5^2} - 2.8 \times 1.7 - 1.7 \times 1.5 - 2.8 \times 1.5)}{\cancel{2.8^2+1.7^2+1.5^2} - 2.8 \times 1.7 - 1.7 \times 1.5 - 2.8 \times 1.5} \\ = 2.8 + 1.7 + 1.5 \\ = 6\end{aligned}$$

$\therefore$ (A) 6

(24) $\frac{x^2 - 25}{(x+5)(x-5)} = ?$

- (A) 0 (B) 1
(C) 5 (D) 7

Solution :

$$\Rightarrow \frac{x^2 - 5^2}{(x+5)(x-5)} = \frac{\cancel{(x+5)} \cancel{(x-5)}}{\cancel{(x+5)} \cancel{(x-5)}} = 1$$

$\therefore$ (B) 1

(25) $\frac{(1.2)^3 + (2.3)^3 + (4.5)^3 - 3 \times 1.2 \times 2.3 \times 4.5}{(1.2)^2 + (2.3)^2 + (4.5)^2 - 1.2 \times 2.3 - 2.3 \times 4.5 - 4.5 \times 1.2}$

- (A) 7 (B) 8 (C) 10 (D) 9

Solution :

$$\Rightarrow a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

પરથી,

$$\Rightarrow \frac{(1.2+2.3+4.5)(\cancel{1.2^2+2.3^2+4.5^2} - \cancel{1.2 \times 2.3} - \cancel{2.3 \times 4.5} - 4.5 \times 1.2)}{(1.2^2 + 2.3^2 + 4.5^2 - \cancel{1.2 \times 2.3} - \cancel{2.3 \times 4.5} - 4.5 \times 1.2)}$$

$= 1.2 + 2.3 + 4.5$

$= 8$

$\therefore$ (B) 8

(26) જો $xy = 18$ અને $x^2 + y^2 = 45$ હોય તો $x + y = ?$

- (A) 36 (B) 18 (C) 27 (D) 9

Solution :

$$\begin{aligned}\Rightarrow (x+y)^2 &= x^2 + y^2 + 2xy \\ (x+y)^2 &= 45 + 2 \times 18 \\ (x+y)^2 &= 45 + 36 \\ (x+y)^2 &= 81\end{aligned}$$

$x + y = \sqrt{81}$

$x + y = 9$

$\therefore$ (D) 9

(27) જો $xy = 36$ અને $x^2 + y^2 = 72$ હોય તો $x + y = ?$

- (A) 10 (B) 11 (C) 12 (D) 14

Solution :

$$\begin{aligned}\Rightarrow (x+y)^2 &= x^2 + y^2 + 2xy \\ (x+y)^2 &= 72 + 2 \times 36 \\ (x+y)^2 &= 72 + 72 \\ (x+y)^2 &= 144\end{aligned}$$

$x + y = \sqrt{144}$

$x + y = 12$

$\therefore$ (C) 12

(28) જો $xy = 28$, $x^2 + y^2 = 81$ તો $x - y = ?$

- (A) 5 (B) 6 (C) 7 (D) 9

Solution :

$$\begin{aligned}\Rightarrow (x-y)^2 &= x^2 + y^2 - 2xy \\ (x-y)^2 &= 81 - 2 \times 28 \\ (x-y)^2 &= 81 - 56 \\ (x-y)^2 &= 25\end{aligned}$$

$(x-y) = \sqrt{25}$

$(x-y) = 5$

$\therefore$ (A) 5

(29) જો $a + b = 3$ અને $ab = -4$ હોય તો $a^3 + b^3 = ?$

- (A) -15 (B) 63 (C) 10 (D) 20

Solution :

$$\begin{aligned}\Rightarrow (a+b)^3 &= a^3 + b^3 + 3ab(a+b) \\ (3)^3 &= a^3 + b^3 + 3 \times (-4) \times 3 \\ 27 &= a^3 + b^3 - 36 \\ a^3 + b^3 &= 27 + 36 \\ a^3 + b^3 &= 63 \\ \therefore (B) 63\end{aligned}$$

(30) $(2x + 3y + 4z)^2 = ?$

- (A) $4x^2 + 9y^2 - 16z^2 - 12xy + 24yz$
(B) $4x^2 + 9y^2 + 16z^2 + 12xy + 24yz + 16zx$
(C) $10x^2 + 12y^2 + 7z^2 - 3xyz$
(D) $4x^2 + 9y^2 + 16z^2 - 3xyz$

Solution :

$$\begin{aligned}\Rightarrow (a+b+c)^2 &= a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \\ \Rightarrow (2x+3y+4z)^2 &= (2x)^2 + (3y)^2 + (4z)^2 + 2 \times 2x \times 3y + 2 \times 3y \times 4z + 2 \times 4z \times 2x \\ &= 4x^2 + 9y^2 + 16z^2 + 12xy + 24yz + 16zx \\ \therefore (B) 4x^2 + 9y^2 + 16z^2 + 12xy + 24yz + 16zx\end{aligned}$$

(31) $\frac{[1.5]^3 + [4.7]^3 + [3.8]^3 - 3 \times 1.5 \times 4.7 \times 3.8}{[1.5]^3 + [4.7]^3 + [3.8]^3 - 1.5 \times 4.7 - 4.7 \times 3.8 - 3.8 \times 1.5} = ?$

- (A) 0 (B) 10 (C) 30 (D) 30

Solution :

$$\begin{aligned}\Rightarrow \frac{a^3 + b^3 + c^3 - 3abc}{a^2 + b^2 + c^2 - ab - bc - ca} \\ = \frac{(a+b+c)(\cancel{a^2 + b^2 + c^2 - ab - bc - ca})}{\cancel{a^2 + b^2 + c^2 - ab - bc - ca}} \\ = a + b + c \\ = 1.5 + 4.7 + 3.8 \\ = 10 \\ \therefore (B) 10\end{aligned}$$

(32) $(x + y)^2 = ?$

- (A) $x^2 + y^2 + 2xy$ (B) $x^2 + y^2 - 2xy$
(C) $x^2 - y^2 - 2xy$ (D) $x^2 - y^2 + 2xy$

Solution :

$$\begin{aligned}\Rightarrow (x+y)^2 &= x^2 + y^2 + 2xy \\ \therefore (A) x^2 + y^2 + 2xy\end{aligned}$$

(33) $(8x - 5)^2 = ?$

- (A) $64x^2 + 8x + 10$ (B) $64x^2 + 80x + 25$
(C) $4x^2 - 20x + 27$ (D) $64x^2 - 80x + 25$

Solution :

$$\begin{aligned}\Rightarrow (a-b)^2 &= a^2 + b^2 - 2ab \\ &= [8x]^2 + [5]^2 - 2 \times 8x \times 5 \\ &= 64x^2 + 25 - 80x \\ &= 64x^2 - 80x + 25 \\ \therefore (D) 64x^2 - 80x + 25\end{aligned}$$

(34) $x + \frac{1}{x} = 2$ ને $x^{100} + \frac{1}{x^{100}} = ?$

- (A) 2 (B) 4 (C) 10 (D) 20

Solution :

$$\begin{aligned}\Rightarrow x \text{ ની બેકી ધાત હોય ત્યારે બીજગણિતિક સૂત્ર મુજબ } \\ \text{જવાબ 2 જ મળે.} \\ \therefore (A) 2\end{aligned}$$

Type # 2

(35) જો $x + y = 41$ તો $(x - 20)^{2021} + (y - 21)^{2021}$ ની કિંમત શોધો.

- (A) 0 (B) 1 (C) 2 (D) 3

Solution :

$$\begin{aligned}\Rightarrow (x-20) + (y-21) &= x + y - 41 \\ &= 41 - 41 \\ &= 0 \\ \therefore (A) 0\end{aligned}$$

(36) $(3x + 2y)^3 + (3x - 2y)^3 = 3kx(3x^2 + 4y^2)$ નો k ની કિંમત શોધો.

- (A) 18 (B) 9 (C) 3 (D) 6

Solution :

$$\begin{aligned}\Rightarrow (3x+2y)^3 + (3x-2y)^3 &= 3kx(3x^2 + 4y^2) \\ &= (3x)^3 + (2y)^3 + 3(3x)(-2y) + (3x)^3 - (2y)^3 - 3(3x)(2y)(3x-2y) = 3kx(3x^2 + 4y^2) \\ \therefore 54x^3 + 72xy^2 &= 9kx^3 + 12kxy^2 \\ \therefore 54x^3 &= 9kx^3 \\ \therefore k &= \frac{54}{9} \\ \therefore k &= 6 \\ \therefore (D) 6\end{aligned}$$